Training Set Analysis

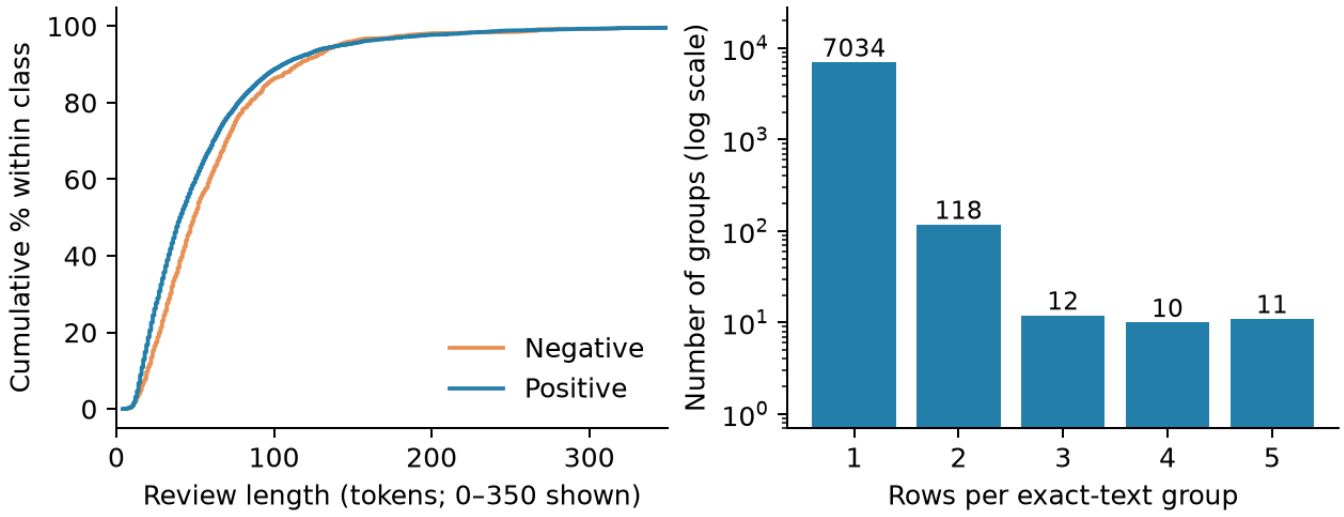
IE4483 | Preparing the shared train/validation split

8 November 2026 | Train-only descriptive analysis and split diagnostics

1. Class structure and review lengths

Only data/train.json is used. The supplied training pool contains **7,401 rows and 2 columns**: one raw-text input and one binary sentiment target. There are no missing/null fields, empty reviews or invalid labels. The official test set is excluded from every statistic and simulation in this report.

Statistic	Negative (0)	Positive (1)
Rows	1,082	6,319
Share of training pool	14.62%	85.38%
Unique exact texts	1,049	6,136
Extra duplicate rows	33	183
Mean length	62.0	55.5
Median length	49.0	41.0
90th percentile	115.9	106.0
95th percentile	140.9	142.1
Maximum length	567.0	1,045.0
Over 150 tokens	42 (3.9%)	282 (4.5%)



Token definition: lowercase text, then regex [a-z']+. Lengths are descriptive counts, not GRU/subword token counts. Percentiles use linear interpolation. Negative reviews have a different length distribution; check length by class after splitting rather than checking class counts alone.

2. Duplicate groups and leakage risk

There are **7,185 unique exact texts** and **216 extra duplicate rows**. Of these groups, 151 contain multiple rows. No exact-text group has conflicting sentiment labels. The largest group contains 5 rows.

Group size (rows)	Number of groups	Rows represented
1	7034	7034
2	118	236
3	12	36
4	10	40
5	11	55

Why ordinary stratification is insufficient

Stratification preserves the label ratio, but independent row assignment can place copies of the same review on both sides. Validation can then reward memorisation. Keep all rows sharing an exact review string together when constructing the shared split.

Diagnostic comparison: 80/20, seed 42

Check	Row-level split	Exact-text grouped split
Training rows	5921	5921
Validation rows	1480	1480
Validation negative / positive	216 / 1264	216 / 1264
Shared exact texts	62	0
Validation rows seen in training	69	0
Mean length: training / validation	56.8 / 55.3	56.9 / 54.7

These are deterministic diagnostic simulations, not a saved or approved shared split. Row simulation: shuffle each class independently with Python Random(42 + label), then take round(20% of rows) for validation. Group simulation: shuffle exact-text groups per class with the same seed rule, then choose the prefix nearest to 20% of that class's rows. This is not sklearn train_test_split and need not reproduce the README's split.

Additional sensitivity check: case-folding and collapsing whitespace gives 7,182 unique groups, 3 fewer than exact matching, with 0 conflicting-label groups. This does not check semantic similarity or punctuation variants; choose and record the grouping policy before finalising the split.

3. Recommended shared split protocol

Recommended starting point: **80% training / 20% validation, seed 42, stratified by sentiment and grouped by review text**. Keep duplicate rows together to preserve the supplied class distribution. Do not silently drop duplicates: that changes sample weighting and should be a separately documented experiment.

Define the grouping policy

Use exact text as the minimum protection. Prefer the conservative case-folded, whitespace-collapsed key if inspection confirms that merged variants are equivalent and no conflicting labels occur. Grouping normalization does not require altering model input text.

Create a deterministic grouped split

Target approximately 216 negative and 1,264 positive validation rows (about 1,480 total). The original 7,401 rows do not divide exactly in an 80/20 ratio, and group sizes can shift counts slightly. Report realised counts rather than requiring an arbitrary exact total.

Save the shared artifact

Export original zero-based row indices, seed, grouping rule, algorithm/version, source SHA-256 and realised counts. All group members must use this one artifact, including the TF-IDF baseline, GRU tuning and evaluation code.

Validate before training

Assert index uniqueness, disjoint train/validation row indices, complete coverage of all rows, both labels present and zero overlap of grouping keys. Compare label ratios, length percentiles by class and long-review rates between partitions. Avoid repeatedly choosing a split based on model performance.

Fit transforms on training only

Build TF-IDF vocabulary, IDF weights or neural vocabulary using the training partition. Derive loss/class weights from that partition. Tune hyperparameters and sequence length using validation; keep test data outside model-selection decisions.

Plan fair evaluation

Report accuracy, macro-F1, per-class precision/recall/F1 and confusion matrix. Validation has only about 216 negative examples, so one missed negative changes negative recall by roughly 0.46 percentage points. Small score differences need cautious interpretation.

Scope and reproducibility

No classifier was trained and no official split was written. Run .venv/bin/python analysis/train_split_report.py to regenerate this PDF, chart and train_split_statistics.json. The JSON records the source-file fingerprint and exact simulation results. Descriptive train-only evidence is sufficient for this preparation; product/reviewer identities and time-based leakage cannot be assessed because those identifiers are absent.